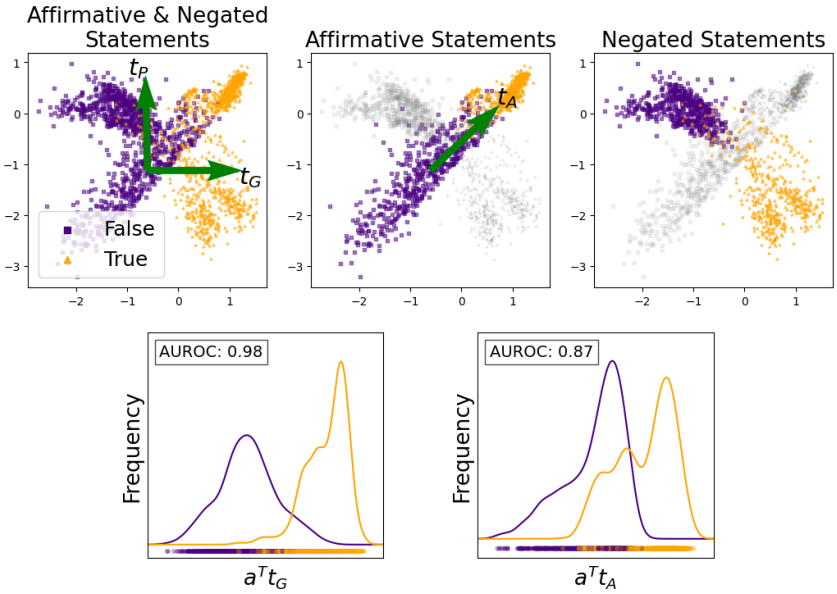

Figure 1: Top left: The activation vectors of multiple statements projected onto the 2D subspace spanned by our estimates for $\mathbf{t}_G$ and $\mathbf{t}_P$. Purple squares correspond to false statements and orange triangles to true statements. Top center: The activation vectors of *affirmative* true and false statements separate along the direction $\mathbf{t}_A$. Top right: However, *negated* true and false statements do not separate along $\mathbf{t}_A$. Bottom: Empirical distribution of activation vectors corresponding to both affirmative and negated statements projected onto $\mathbf{t}_G$ and $\mathbf{t}_A$, respectively. Both affirmative and negated statements separate well along the direction $\mathbf{t}_G$ proposed in this work.

Polarity Direction), a new method for LLM lie detection which classifies statements as true or false. Through empirical validation that extends beyond the scope of previous studies, we show that TTPD can accurately distinguish true from false statements under a broad range of conditions, including settings not encountered during training. In real-world scenarios where the LLM itself generates lies after receiving some preliminary context, TTPD can accurately detect this with 94% accuracy, despite being trained only on the activations of simple factual statements. We compare TTPD with three state-of-the-art methods: Contrast Consistent Search (CCS) by Burns et al. [2023], Mass Mean (MM) probing by Marks and Tegmark [2023] and Logistic Regression (LR) as used by Burns et al. [2023], Li et al. [2024] and Marks and Tegmark [2023]. Empirically, TTPD achieves the highest generalization accuracy on unseen types of statements and real-world lies and performs comparably to LR on statements which are about unseen topics but similar in form to the training data.

3. **Universality across model families:** This internal two-dimensional representation of truth is remarkably *universal* [Olah et al., 2020], appearing in LLMs from different model families and of various sizes. We focus on the instruction-fine-tuned version of LLaMA3-8B [AI@Meta, 2024] in the main text. In Appendix G, we demonstrate that a similar two-dimensional truth subspace appears in Gemma-7B-Instruct [Gemma Team et al., 2024a], Gemma-2-27B-Instruct [Gemma Team et al., 2024b], LLaMA2-13B-chat [Touvron et al., 2023], Mistral-7B-Instruct-v0.3 [Jiang et al., 2023] and the LLaMA3-8B base model. This finding supports the Platonic Representation Hypothesis proposed by Huh et al. [2024] and the Natural Abstraction Hypothesis by Wentworth [2021], which suggest that representations in advanced AI models are converging.

The code and datasets for replicating the experiments can be found at `https://github.com/sciai-lab/Truth_is_Universal`.

After recent studies have cast doubt on the possibility of robust lie detection in LLMs, our work offers a remedy by identifying two distinct "truth directions" within these models. This discovery explains the generalisation failures observed in previous studies and leads to the development of a